

### Advanced (Habanero)

- Q1.** Which function represents a vertical shift of  $f(x) = x^2$  by 3 units down?  
(A)  $f(x) = x^2 - 3$   
(B)  $f(x) = x^2 + 3$   
(C)  $f(x) = (x - 3)^2$   
(D)  $f(x) = (x + 3)^2$   
(E)  $f(x) = -x^2$
- Q2.** What is the equation of the function  $f(x) = x^2$  after a horizontal shift of 2 units to the left?  
(A)  $f(x) = (x + 2)^2$   
(B)  $f(x) = (x - 2)^2$   
(C)  $f(x) = x^2 + 2$   
(D)  $f(x) = x^2 - 2$   
(E)  $f(x) = -x^2$
- Q3.** The function  $f(x) = x^2$  is reflected over the x-axis. What is the new equation?  
(A)  $f(x) = -x^2$   
(B)  $f(x) = x^2$   
(C)  $f(x) = -x$   
(D)  $f(x) = x$   
(E)  $f(x) = x^3$
- Q4.** What is the equation of  $f(x) = x^2$  after a reflection over the y-axis?  
(A)  $f(x) = x^2$   
(B)  $f(x) = -x^2$   
(C)  $f(x) = (-x)^2$   
(D)  $f(x) = -(-x)^2$   
(E)  $f(x) = x$
- Q5.** The function  $f(x) = x^2$  is stretched vertically by a factor of 2. What is the new equation?  
(A)  $f(x) = 2x^2$   
(B)  $f(x) = x^2/2$   
(C)  $f(x) = x^2 + 2$   
(D)  $f(x) = x^2 - 2$   
(E)  $f(x) = -x^2$
- Q6.** What is the equation of the function  $f(x) = x^2$  after a horizontal compression by a factor of 2?  
(A)  $f(x) = (2x)^2$   
(B)  $f(x) = (x/2)^2$   
(C)  $f(x) = x^2 + 2$   
(D)  $f(x) = x^2 - 2$   
(E)  $f(x) = -x^2$
- Q7.** The function  $f(x) = x^2$  is shifted 1 unit to the right and 2 units down. What is the new equation?  
(A)  $f(x) = (x - 1)^2 - 2$   
(B)  $f(x) = (x + 1)^2 + 2$   
(C)  $f(x) = x^2 - 2$   
(D)  $f(x) = x^2 + 2$   
(E)  $f(x) = -x^2$
- Q8.** Which of the following is the graph of  $f(x) = -x^2$ ?  
(A) A parabola that opens up  
(B) A parabola that opens down  
(C) A horizontal line  
(D) A vertical line  
(E) A circle

#### Open Ended Questions (FRQ)

- Q9.** Graph the function  $f(x) = x^2$  and its transformation  $f(x) = 2(x + 1)^2 - 3$ . Label the vertex of each graph.
- Q10.** Suppose  $f(x) = x^2$  and  $g(x) = f(2x)$ . Describe the transformation from  $f(x)$  to  $g(x)$  and graph both functions on the same coordinate plane.

#### Chef's Tips